

CoAS: Composite Audio Steganography Based on Text and Speech Synthesis

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Abstract—Digital steganography is the practice of embedding secret information in original normal data to enable covert communication. With the rapid advancement of generative models, generative steganography has gained renewed vitality. As a key medium on the Internet, audio has also become a focus of steganographic research. However, existing audio steganography methods rely on traditional audio synthesis models, which often suffer from suboptimal synthesis quality. In contrast, diffusion models perform well in audio synthesis tasks, but there is a lack of targeted secure audio steganography methods based on them. In addition, existing steganography schemes are generally limited to transmitting only the steganographic object, and other key elements need to be negotiated in advance, which limits their practicality. To address these issues, we propose CoAS, a composite audio steganography method based on text and speech synthesis. Firstly, we use a provably secure linguistic steganography method to embed the synchronous side information required for audio steganography, and then replace the gaussian noise in the diffusion models with message-driven sampling during the audio generation process. Both theoretical analysis and experimental results validate the security and practicality of our composite steganography method in the real world. Audio samples are available at <https://material.github.io/coas.github.io>

Index Terms—Steganography, provably secure, text, audio, diffusion model.

I. INTRODUCTION

DIGITAL steganography is the practice of hiding secret messages in data media (e.g., images, video, audio, text, etc.) to hide the message from detection by third-party observers. The sender hides the secret message inside the cover, creates the steganographic object, and transmits it to the receiver, who extracts the secret message from the steganographic object. Throughout the process, the third-party analysts use steganalysis tools that attempt to distinguish between cover and stego objects.

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There are different perspectives for defining steganographic security. In 1998, Cachin [1] proposed a theoretical definition of steganographic security using the concept of relative entropy in information theory. This definition uses the relative entropy between the cover and stego object as a measure of steganographic security and generally speaking, a steganographic system is said to be ϵ -secure if the relative entropy between the cover and stego object is less than a small value ϵ , which implies that for third-party analysts, it is difficult to distinguish between cover and stego objects. Hopper et al. [2] defined steganographic security from a computational security point of view, proposing that steganography is considered computationally feasible and secure if the detection advantage of an attacker is negligible compared with the security parameter under polynomial time conditions.

With Cachin proposing a definition of steganographic security, several steganographic algorithms related to the distribution of cover-stego have subsequently been proposed [2], [3], [4]. These include rejection sampling-based algorithms, in which cover samples are repeatedly drawn until a sample whose hash value matches that of the encrypted secret message is obtained. Encryption algorithms (e.g., XOR between a secret message and a shared key) are shared between the sender and receiver. Hopper et al. [2] improved the above method based on the adequacy of the entropy of the data distribution and the assumption of repeatability based on historical samples and extended the rejection sampling method to arbitrary distributions. Reyzin and Russell [3] further enhanced the robustness and steganographic efficiency of the method from Hopper. Von Ahn and Hopper [4] devised a steganography method based on public keys and proposed the chosen steganographic text attack.

Owing to the lack of precise sampling covers in traditional data environments, it has been challenging to meet the requirements for provably secure steganography. Consequently, empirical steganography has rapidly developed in recent times, especially content-adaptive steganography methods. For typical steganographic medium audio, high-security steganographic methods are formed by leveraging defined steganographic distortion functions and steganographic codes that approximate the rate-distortion bounds (such as STC [5], SPC [6], and LDGM codes [7]). It was not until Chen et al. [8] established a connection between provably secure steganography and generative models that the academic focus in this field shifted back to provably secure steganography. In the

audio domain, they proposed a provably secure steganographic method based on the speech synthesis models Waveglow [9] and WaveNet [10]. However, these two audio-generative models are outdated. Currently, the diffusion models [11], [12] have become the most important generative models in the field of audio generation, surpassing Waveglow and WaveNet in terms of audio quality. Moreover, owing to the inherent differences in network structure between generative models, previous audio steganography methods cannot be directly applied to diffusion models. Therefore, the development of secure steganography methods for audio diffusion models remains an unsolved challenge.

In addition, existing steganographic algorithms [4], [13], [14] are usually limited to a priori data distribution knowledge and require the sender and receiver to share some key information before steganographic transmission. The GAN-based steganography methods [15], [16] require the generation process of models with the shared key to ensure correct message extraction. When the attacker uses these steganographic models to randomly generate stegos and check the results of generation, he will find that there is a loss of diversity. Under such a threat model, the concealment of steganographic behavior cannot be guaranteed. We will describe this threat model in detail in Section III-A.

To address these issues, we propose an ϵ -secure Composite Audio Steganography method (CoAS) based on text and speech synthesis with diffusion models. For covert communication, it requires that the sender and receiver share elements, such as the random seed and model generation parameters. Leveraging another channel for parameter transmission will undoubtedly introduce the risk of exposure to covert communication behavior. In CoAS, we exploit the multicover nature of audio to address this issue. Specifically, we employ existing provably secure linguistic steganography methods to embed the random seed into the corresponding audio text, enabling simultaneous transmission of the stego audio and parameters.

For the audio steganography part, we analyzed the forward denoising process of the audio diffusion model. In the audio generation process, the diffusion model continuously removes Gaussian noise from the initial noise to obtain synthetic speech. We replace the Gaussian noise in the penultimate step of the denoising process with the Gaussian vector sampled driven by the secret message, resulting in steganographic audio. As long as the receiver can synchronize the random seed hidden in the audio text with the sender, the receiver can reproduce part of the audio generation process. Then, with the received audio sequence, we can obtain the Gaussian vector, thereby mapping back to the secret message. We introduce a message mapping method to map a uniformly distributed secret message to a Gaussian distribution. The Gaussian noise obtained through message-driven sampling is indistinguishable from that obtained by random sampling. Therefore, the distribution of the stego is very close to that of the cover audio. The only difference is that we omit the final denoising step for message extraction. We also theoretically analyzed that as the number of denoising steps increases, the distribution distance between the cover and stego audio decreases, approaching zero.

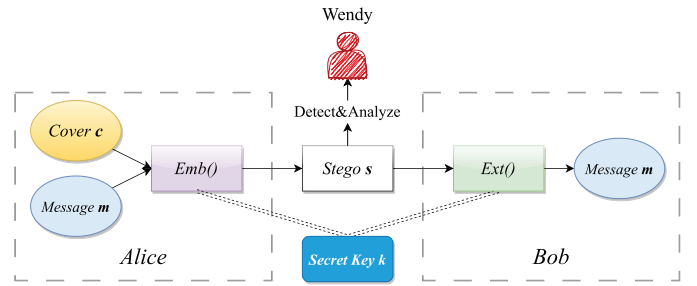


Fig. 1. The classical Prisoners' Problem in steganography.

The contributions of this work can be summarized as follows:

- We take advantage of the multicover nature of audio and propose CoAS, a method of composite audio steganography based on text and speech synthesis. In addition to audio steganography, we utilize provably secure linguistic steganography methods to hide key parameters of audio steganography in the audio text. We achieve an ϵ -secure steganography method while maintaining the diversity of generated audio.
- We explored the sampling process of the audio diffusion model and introduced a message mapping method to replace the noise randomly sampled by the generative model with the noise obtained through message-driven sampling, thereby achieving efficient embedding of the secret message.
- We theoretically prove the ϵ -security of the CoAS, and the steganalysis experiment further verifies the security of the method. The audio quality metrics show that this steganographic framework has no negative impact on the quality of the audio generated by CoAS, which is still better than the traditional audio steganography method.

II. RELATED WORK

A. The Prisoners' Problem

Steganography, the practice of concealing secret messages within innocuous content, was formally conceptualized by Simmons through the "Prisoners' Problem" model [17]. The model revolves around the incarceration of two prisoners, Alice and Bob, who are confined to separate rooms within the prison as shown in Fig. 1. They conspired to escape together. Despite the consent from the prison guard, Wendy, for Alice and Bob to communicate through a public channel, all their communications undergo scrutiny by Wendy. Any abnormality detected in their communication prompts Wendy to terminate it immediately.

According to the Prisoners' Problem, a steganographic scheme can be regarded as a pair of embedding and extraction functions $\text{Emb}()$ and $\text{Ext}()$ for Alice and Bob, respectively [18].

$$\begin{aligned} \text{Emb}(c, k, m) &= s, \\ \text{Ext}(s, k) &= m, \end{aligned} \quad (1)$$

where c, k, m, s are the cover object, secret key, message, and stego object, respectively. Wendy judges whether the object s

is innocent or not by all the possible knowledge except the secret key k , according to Kerckhoff's principle [19].

B. The Definition of Steganographic Security in Theory

The information-theoretic definition of steganographic security was proposed by Cachin [1]. Assuming that the cover is extracted from the set $\mathcal{C}$ according to the probability distribution P_C , the steganographic method generates a stego with the distribution P_S . The distance between these two distributions can be measured via the relative entropy:

$$D(P_C||P_S) = \sum_{x \in \mathcal{C}} P_C(x) \log \frac{P_C(x)}{P_S(x)}. \quad (2)$$

When $D(P_C||P_S) \leq \epsilon$, the steganographic system is said to be ϵ -secure. Here, ϵ represents a small constant to evaluate the upper bound of the difference.

C. Audio Steganography

Traditional audio steganography can be implemented in three domains: the temporal domain, frequency domain, and wavelet domain [20]. In the temporal domain, the most common is Least Significant Bit (LSB) steganography [21], whose basic idea is to replace the least significant bit in audio sample values with a secret bit. Following the LSB steganography algorithm, optimizations have been made in terms of security, embedding efficiency, and steganographic capacity. Other typical audio steganography methods in this domain include echo steganography [22], which uses the masking effect of the human auditory system. By introducing a short echo to the host signal, this method embeds the data into an audio signal. The stego audio retains the same statistical and perceptual characteristics as the cover. In the frequency domain, the methods use the masking effect of the human auditory system, which includes spread spectrum hiding, phase coding, etc. The spread spectrum is developed in communications to ensure proper recovery of a signal sent over a noisy channel by producing redundant copies of the data signal. The spread spectrum technique spreads hidden signal data through the frequency spectrum. The phase coding [23] method takes advantage of the insensitivity of the human ear to the relative phase of different spectral components. The data is hidden in selected phase components of the original speech spectrum. In the wavelet domain, Sheikhan et al. [24] uses the discrete wavelet transform, which can decompose the signal into high-frequency and low-frequency parts. The secret information can be embedded into discrete wavelet coefficients by combining the wavelet energy, masking effect, and adaptive LSB. With powerful steganographic codes, some adaptive audio steganographic methods [25], [26] have been proposed, which largely improve security.

With the development of generative models, many generative audio steganography methods have emerged recently. Jiang et al. [15] applied the short-time Fourier transform to concatenate the cover and stego audio spectrograms with the smartGAN model in the encoder, and utilized a CNN-based steganalyzer [27] as the discriminator. Chen et al. [16] utilized a GAN to produce high-quality cover speech for

steganography and embedded secret messages in a manner that maximizes the misclassification of corresponding stego speech by steganalysis as the cover. Moorthy and Venkataraman [28] proposed a GAN-based approach to embed stego audio into cover audio in the temporal domain. Their model employed a U-Net-based generator, an LSB-based embedder, and a discriminator to achieve good imperceptibility of the stego audio.

Although GAN-based audio steganography is technically feasible, embedding secret messages into clear cover audio inevitably leaves traces of covert communication. In response to this issue, Chen et al. [8] utilized text-to-speech generative models such as Waveglow [9] to simultaneously generate cover audio and stego audio. A positive aspect of the stegosystem based on Waveglow is that it can maintain the same distribution between the stego and the cover audio in WAV format, resulting in high concealment and good undetectability. However, limited by the performance of Waveglow, the audio generated falls short in quality.

Nowadays, the diffusion model greatly improves the audio quality. Different from previous studies, we chose the diffusion model to generate the stego audio, ensuring high-quality audio generation to alleviate suspicion from steganalysis.

D. Diffusion Model

Denoising Diffusion Probabilistic Models [12], [29], [30] are emerging generative models achieving excellent results in benchmark generation tasks [31] and have proven their ability to produce high-quality samples. The basic idea of DDPMs is to train a denoising network to reverse the diffusion process.

Given data sampled from a real data distribution $q(\mathbf{x}_0)$, the diffusion process adds a small amount of Gaussian noise to $\mathbf{x}_0$ in T steps, produced by a fixed Markov chain. The noise chain is controlled by a variance schedule $\{\beta_t \in (0, 1)\}_{t=1}^T$. For a small positive constant β_t , a small Gaussian noise is added from $\mathbf{x}_{t-1}$ to the distribution of $\mathbf{x}_t$ under the function $q(\mathbf{x}_t|\mathbf{x}_{t-1})$.

$$q(\mathbf{x}_{1:T}|\mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1}),$$

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}). \quad (3)$$

The corresponding reverse process is also a Markov chain from $\mathbf{x}_T$ to $\mathbf{x}_0$ parameterized by a shared diffusion model θ , which aims to recover samples from Gaussian noise by eliminating the Gaussian noise added in the diffusion process in each iteration.

Diffusion models have been proven [31] to learn diverse data distributions across multiple domains, such as images and audio, and achieve leading performances in many generative tasks. Taking the audio generation task as an example, the diffusion model has made great strides compared with the previous autoregressive model, WaveNet, and the flow-based model, Waveglow, in terms of audio quality, among other things. In the crowd-sourced 5-scale MOS [32] tests, the diffusion model FastDiff [33] scores slightly better than WaveNet does, but the time required to generate one second of audio is reduced from 85s to 0.02s, whereas the Waveglow model

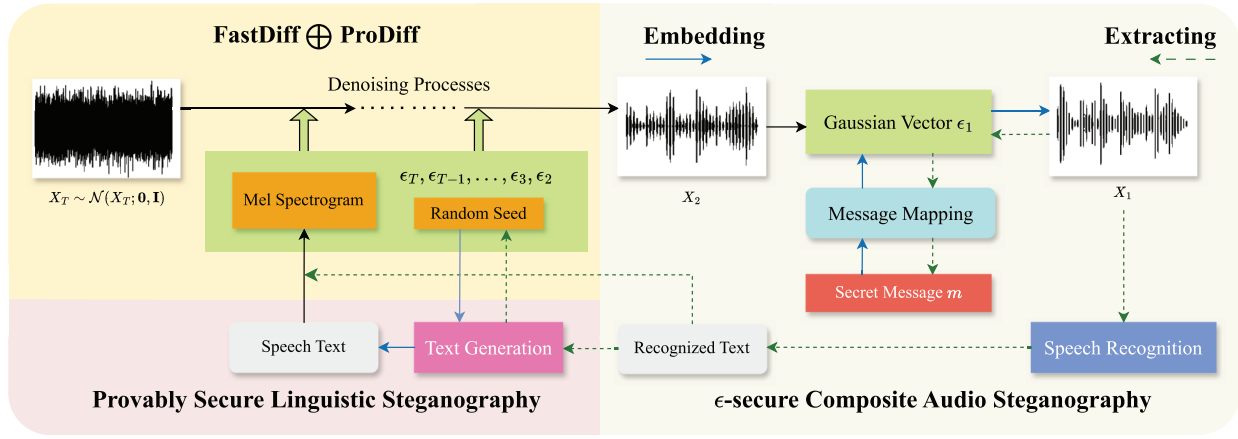


Fig. 2. CoAS: The composite audio steganographic scheme based on diffusion models. The Gaussian vector obtained from the secret message mapping is applied to the denoising process of the diffusion model, and the audio becomes the cover of the secret message. At the same time, the shared parameters required by the above process are embedded in the text corresponding to the audio and guide the generation process.

scores are significantly lower than the score of the diffusion model by 0.4 points.

III. CoAS: COMPOSITE AUDIO STEGANOGRAPHY

In this section, we introduce CoAS, a composite audio steganography method based on text and speech synthesis. First, we introduce the definition of the threat model in this paper. Then, we introduce the proposed CoAS with diffusion models as the core and the parameter synchronization technology used in it. Finally, we prove the ϵ -security of the CoAS. Each module in CoAS is further discussed in the following sections.

A. Threat Model

Before introducing the CoAS steganography system, let us first briefly introduce the threat model of the generative steganography system.

According to Kerckhoff's principle, the model used by the steganographic system needs to be public. At the same time, the attacker has black box access to the modified steganographic model and can repeatedly generate different stegos for inspection. An attacker $\mathcal{A}$ can use any initial conditions to drive the generation process and generate stegos X_s , similar to the *chosen hidden text attacks* [2], which we refer to as *chosen stego attacks*.

The generative steganography method mentioned in Section II-C requires the sender to fix the random seed in the model to achieve correct message extraction, that is, $X = G(I, s, m)$. Among them, I represents the initial condition of the input generation model, G represents the generation model used by the steganography system, m represents the secret message to be embedded, and s is the random seed. After such models are public, the attacker can carry out *chosen stego attacks* and generate a series of results $\{X_k = G(I, s, \emptyset), k = 1, 2, 3, \dots\}$ with the same I .

B. Hiding Messages in Diffusion Process

To achieve high-quality audio synthesis to enhance the security of the steganography algorithm, we chose FastDiff

[33] and ProDiff [34] as the generators to produce stego audio in the CoAS steganography system.

The FastDiff employs a stack of time-aware location-variable convolutions with diverse receptive field patterns to model long-term time dependencies with adaptive conditions. A noise predictor is included and optimized in the framework for reducing reverse iterations while maintaining audio quality. The ProDiff requires only 2 iterations to synthesize high-fidelity mel-spectrograms, while it maintains sample quality and diversity competitive. The excellent generation of diffusion models provides a high-quality carrier for generative audio steganography.

These models enable text-to-speech conversion at low computational cost. The FastDiff initially samples the noisy audio, denoted as x_t , from a standard Gaussian distribution, i.e., $x_t \sim \mathcal{N}(x_t; 0, I)$, and then progressively denoises to obtain the audio sequence. ProDiff synthesizes high-fidelity mel-spectrograms from text inputs. These mel-spectrograms will be used as conditions to guide audio generation in the denoising process of the FastDiff model. As shown in Fig. 2, FastDiff takes the mel-spectrogram converted from text by ProDiff and a randomly sampled Gaussian noise sequence, ϵ_t , as inputs. Then, the model removes noise from the initial Gaussian noise x_t to yield a high-quality audio sequence x_0 . The denoising process follows the equations below:

$$x_{t-1} = \frac{1}{\sqrt{1-\beta_{t-1}}} \left(x_t - \frac{\beta_{t-1}}{\sqrt{1-\alpha_{t-1}^2}} \cdot \theta(x_t) \right) + \sigma_t \epsilon_{t-1},$$

$$x_0 = \frac{1}{\sqrt{1-\beta_0}} \left(x_1 - \frac{\beta_0}{\sqrt{1-\alpha_0^2}} \cdot \theta(x_1) \right), \quad (4)$$

where α and σ are parameter sequences derived from the noise schedule β . The value of β is pre-set and automatically selected by the diffusion model based on the denoising iteration step t , during the audio generation process. Each

parameter can be calculated via the following equation:

$$\alpha_i^2 = \prod_{s=1}^i (1 - \beta_s),$$

$$\sigma_i^2 = \beta_i \cdot \frac{1 - \alpha_{i-1}}{1 - \alpha_i}. \quad (5)$$

The FastDiff model θ takes the denoised result $\mathbf{x}_i$ from the previous step and the mel-spectrogram as input. The initial noisy audio $\mathbf{x}_t$ and Gaussian noise sequence $\epsilon_i (i = 1, 2, \dots, t-1)$ in the generation process are the key components of randomness in this algorithm, providing the potential for message embedding. Since the last step in the generation process, which generates $\mathbf{x}_0$, does not contain any information about noise sequence ϵ_i , we have to cancel this step in the process and take $\mathbf{x}_1$ as the audio sequence.

For message embedding, the CoAS replaces ϵ_1 of the Gaussian noise sequence with message-based noise, so that the message can be coupled into the audio generation process. In detail, we map the secret message $\mathbf{m}$ into a Gaussian vector $\mathbf{z}$, maintaining the same distribution as the noise sequence, via the message mapping method. A detailed description of the message mapping method is given in Section III-C. Then CoAS replaces the ϵ_1 with $\mathbf{z}$. After this, the sender can complete the entire generation process as described in Eq. (4) to obtain the stego audio $\mathbf{x}_1$ and send it to the receiver.

Considering that a random number seed can control the initial noisy audio $\mathbf{x}_t$ and Gaussian noise sequence $\epsilon_i (i = 2, \dots, t-1)$, this opens up the possibility for the receiver to reconstruct the generation process and extract the secret message correctly. When the sender and receiver share the random seed and model parameters θ, β , etc., the receiver can run part of the generation process in Eq. (4) to obtain the same intermediate sequence $\mathbf{x}_2$, and accurately extract the Gaussian vector $\mathbf{z}$ from the stego audio $\mathbf{x}_1$ according to the following equation:

$$\mathbf{z} = \frac{1}{\sigma_1} \left[\frac{1}{\sqrt{1 - \beta_1}} \left(\mathbf{x}_2 - \frac{\beta_1}{\sqrt{1 - \alpha_1^2}} \cdot \theta(\mathbf{x}_2) \right) - \mathbf{x}_1 \right]. \quad (6)$$

C. Message Mapping

Inspired by existing provably secure steganography methods [8], [35], [36], we introduce a reversible message mapping method from uniform distribution to Gaussian noise. Generally, for steganography, secret messages are usually pre-encrypted with symmetric cryptography, resulting in a random binary bit sequence for the steganographic process. A straightforward example of this is to perform an XOR operation between the message $\mathbf{m}$ and the key $\mathbf{k}$.

$$\mathbf{m}_e = \text{Enc}(\mathbf{m}, \mathbf{k}), \text{e.g., } \mathbf{m}_e = \mathbf{m} \oplus \mathbf{k}. \quad (7)$$

Regarding the XOR above encryption, if the key is of the same length as the message and is generated randomly, the encrypted message $\mathbf{m}_e$ can be viewed as a uniformly distributed variable. Cryptographically secure random number generators are computationally indistinguishable from true randomness. The same principle applies to other symmetric

encryption algorithms that have been proven to be computationally secure. Therefore, the message mapping method required for CoAS is mapping from the uniform distribution of $\mathbf{m}_e$ to a Gaussian distribution of vector $\mathbf{z}$.

First, we define the payload p in terms of the number of message bits carried by each dimension of the vector $\mathbf{z}$. For a given s -bit encrypted message $\mathbf{m}$ (for simplicity, $\mathbf{m}$ is used in place of $\mathbf{m}_e$), we map $\mathbf{m}$ to $\mathbf{z}$ via the following mapping method.

Take the p bits from the secret message as a segment, denoted as $m_1, m_2, \dots, m_p$, and convert this group of binary numbers to a decimal fraction r in the interval $[0, 1)$:

$$r = \frac{1}{2^p} \left(\sum_{i=1}^p m_i \cdot 2^{i-1} + \omega \right), \quad (8)$$

where $\omega = \text{rand}(0, 1)$, is to ensure that the obtained decimal fraction can be correctly mapped to the Gaussian distribution.

We use the following function to map r to one component z_j of the Gaussian vector $\mathbf{z}$:

$$z_j = F^{-1}(r). \quad (9)$$

Algorithm 1 Mapping Function $\mathcal{M}$

Require: Payload p , Encrypted message $\mathbf{m}$, CDF for Gaussian distribution F

Ensure: Gaussian vector $\mathbf{z}$

```

1: for  $i$  in range  $(0, \text{len}(\mathbf{m}), p)$  do
2:   Let  $\mathbf{y} = [m_{i+1}, m_{i+2}, \dots, m_{i+p}]$ 
3:   Define  $\mathbf{y} \rightarrow [y_1, y_2, \dots, y_p]$ 
4:    $\omega = \text{rand}(0, 1)$ 
5:    $r = \frac{1}{2^p} \left( \sum_{j=1}^p y_j \cdot 2^{p-j} + \omega \right)$ 
6:    $z_i = F^{-1}(r)$ 
7:    $\mathbf{z} = \mathbf{z} \parallel z_i$ 
8: end for
9: return  $\mathbf{z}$ 
```

The range of j , which corresponds to the dimensions of the Gaussian vector $\mathbf{z}$, is determined by the message length s and the payload p , with a maximum value of $\lceil \frac{s}{p} \rceil$. The function F^{-1} is the inverse of the Cumulative Distribution Function (CDF) of the Gaussian distribution. This is why we add ω to Eq. (8): the function F^{-1} cannot accept $r = 0$ as input. This way, a segment of the message with a length of p is mapped to the value z_j that conforms to a Gaussian distribution. Details of the mapping process are given in Algorithm 1.

Consider the process the receiver extracts the secret message from the Gaussian vector $\mathbf{z}$ through inverse mapping: First, the receiver needs to compute the value $r = F(z_j)$ of the Gaussian vector component z_j in the CDF of the Gaussian distribution. Then find the interval $(\frac{n}{2^p}, \frac{n+1}{2^p})$ where r is. The value of n can be calculated directly from $n = \lfloor r \cdot 2^p \rfloor$. Afterward, consider n as the decimal representation of a segment, which can be converted to a binary representation to recover p bits of the secret message. Repeat the process until the s -bit message has been fully recovered. Details of the inverse mapping process are given in Algorithm 2.

Fig. 3 shows an example of a complete message mapping and reverse mapping process of payload $p = 3$.

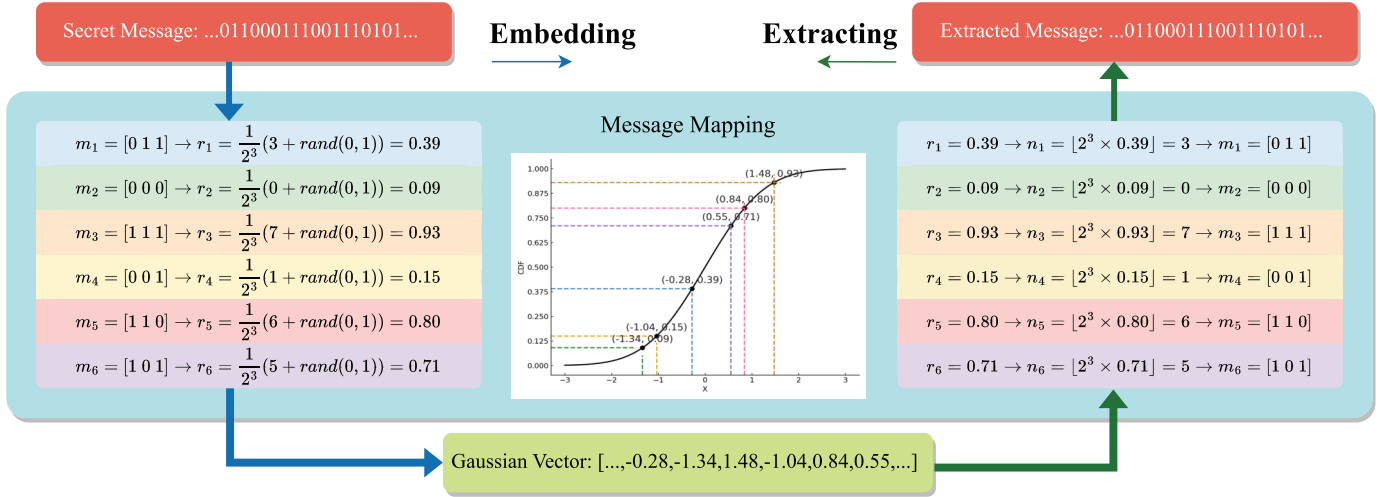


Fig. 3. Message mapping: During the message embedding process, the secret message is firstly divided into a set of segments s_i according to the value of payload p (with $p = 3$ in the figure), and according to Algorithm 1, the components of the corresponding Gaussian vector $\mathbf{z}$ are calculated and obtained. Finally, the Gaussian vector $\mathbf{z}$ is applied to the generation process of the stego audio. On the other hand, in the extraction process, the receiver, after extracting the same Gaussian vector from the stego audio, can obtain the same message segments via Algorithm 2. From this, the secret message can be entirely extracted.

Algorithm 2 Inverse of Mapping Function $\mathcal{M}$

Require: Payload p , Gaussian vector $\mathbf{z}$, CDF for Gaussian distribution F

Ensure: Encrypted message $\mathbf{m}$

- 1: **for** each item z_j in $\mathbf{z}$ **do**
- 2: $r = F(z_j)$
- 3: $n = \lfloor r \cdot 2^p \rfloor$
- 4: $\mathbf{m}_j = \text{Binary}(n)$
- 5: $\mathbf{m} = \mathbf{m} \parallel \mathbf{m}_j$
- 6: **end for**
- 7: **return** $\mathbf{m}$

D. Parameter Synchronization via Provably Secure Linguistic Steganography

In the above message embedding algorithm, the sender and receiver need to share a random number seed to fix the initial noisy audio $\mathbf{x}_t$ and the Gaussian noise sequence ϵ_i ($i = 1, 2, \dots, t-1$) when generating the stego audio.

According to the threat model described in Section III-A, when an attacker $\mathcal{A}$ conducts the *chosen stego attack* on the above audio steganography method, the series of generated results $\{X_k = G(I, s, \emptyset), k = 1, 2, 3, \dots\}$ will be exactly the same, due to the repeated use of random number seed. This phenomenon directly leads to the loss of diversity of the audio generation model and the exposure of the steganographic behavior. Updating the random number seed from other channels is also undesirable, as it will arouse the suspicion of the attacker.

According to the above descriptions, we need to update the random seed while transmitting the audio file without affecting the steganography security. We find that audio files have the nature of multicover and we can recover the text corresponding to the audio through speech recognition. Therefore, we can use the provably secure linguistic steganography method to hide the random number seed in the audio text. To achieve the

above function, we designed the composite audio steganography based on text and speech synthesis.

In this paper, we take Discop [37], a provably secure linguistic steganography, as an example to illustrate the operation process of the CoAS steganography system. Discop is a provably secure linguistic steganography method based on ‘distributed copies’. It constructs multiple distributed copies during the text generation process. When the large language model performs random sampling to determine the next token, the sender uses the secret message as the index of the distribution to choose which distribution to sample from. The sampling value of the large language model can be fixed and synchronized to the receiver through the random number seed. The receiver can reproduce the text generation process and derive the distribution index based on the sampled token results and the synchronized sampling value, and then extract the secret message. Discop does not change the probability distribution of text sampling, so the security is reduced to that of a cryptographically secure pseudo-random number generator (CSPRNG).

Theoretically, Discop can be deployed on any explicit generative model that can yield probability distributions. Given that embedding a random number seed into text is suitable for text generation tasks, we choose LLaMA-2 [38] as the generative model. Besides, to help both parties in communication eliminate token segmentation ambiguity in Discop, we used the SyncPool [39] to optimize the text generation process. We directly employ the pre-trained models provided by Hugging Face [40].

When both the sender and receiver share the model and steganography algorithm, the sender first selects several common words as the initial context. These brief words will be used in the Discop embedding algorithm to write a continuation with the random seed embedded. The continuation process is shown in Fig. 4. Then, the sender inputs the audio text into FastDiff+ProDiff to generate the corresponding audio file as stego audio and sends it to the receiver.

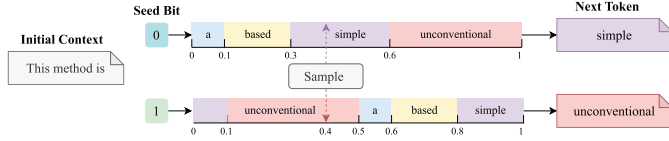


Fig. 4. Discop: The sender selects the sampling results in different distribution copies according to the random seed bit without changing the locations where the explicit generative model samples, thus obtaining a specific continuation of the initial context. The receiver can replicate the same continuation process and, based on the audio text, find the corresponding distribution copies to extract the same random seed bit by bit.

After receiving the audio, the receiver can obtain the same text through speech recognition. The receiver, having the full audio text, can reproduce the text generation process of the Discop based on the same initial context. Compare each sample result with the full audio text to determine which distribution copy was chosen by the sender, and extract the embedded random number seed bit by bit. After that, following the audio embedding and extraction method described above, the sender and receiver can successfully share a random number seed, which is used to extract the Gaussian vector $\mathbf{z}$, and then obtain the secret message $\mathbf{m}$ through the inverse mapping process of CoAS.

It should be noted that the linguistic steganography method suitable for the CoAS is not limited to Discop, and this system can use any provably secure linguistic steganography method [41], [42]. Given the advantages of large language models in generating text quality, the audio generation quality of the CoAS system will be better without affecting security.

E. Discussion of the ϵ -Security

In this section, we discuss the ϵ -security of the CoAS from two aspects. On the one hand, we calculate the KL-Divergence between the cover and the stego audio from the perspective of steganographic security defined based on information theory. On the other hand, based on the threat model introduced in Section III-A, we discuss the security performance of the CoAS under it.

1) *KL-Divergence*: Based on the definition of ϵ -security, we discuss the security of the entire steganographic system in terms of the KL-Divergence between the cover and stego audio generated by the diffusion models. First, we set the distribution of cover audio $\mathbf{x}$ as $P_C(\mathbf{x})$ and the stego as $P_S(\mathbf{x})$. The Gaussian vectors used in the last step of the generation process for both are ϵ_c and ϵ_s , respectively.

If the random number seed and model parameters θ are shared between the sender and receiver, both can generate the same intermediate sequence $\mathbf{x}_2$. We denote $\mathbf{x}_2 \sim Q(\mathbf{x}_2)$, and from this, we begin to compute the distributions P_C and P_S according to Eq. (4).

$$\begin{aligned} P_C(\mathbf{x}) &= \frac{1}{\sqrt{1-\beta_1}} \left[Q(\mathbf{x}_2) - \frac{\beta_1}{\sqrt{1-\alpha_1^2}} \cdot \theta(Q(\mathbf{x}_2)) \right] + \sigma_1 \epsilon_c, \\ P_S(\mathbf{x}) &= \frac{1}{\sqrt{1-\beta_1}} \left[Q(\mathbf{x}_2) - \frac{\beta_1}{\sqrt{1-\alpha_1^2}} \cdot \theta(Q(\mathbf{x}_2)) \right] + \sigma_1 \epsilon_s. \end{aligned} \quad (10)$$

Then we calculate the theoretical KL-Divergence as follows:

$$\begin{aligned} KL[P_C \| P_S] &= E_{\mathbf{x} \sim P_C(\mathbf{x})} \left[\log \frac{P_C(\mathbf{x})}{P_S(\mathbf{x})} \right] \\ &= E_{\mathbf{x} \sim P_C(\mathbf{x})} \left[\log \left(1 + \frac{P_C(\mathbf{x}) - P_S(\mathbf{x})}{P_S(\mathbf{x})} \right) \right] \\ &= E_{\mathbf{x} \sim P_C(\mathbf{x})} \left[\log \left(1 + \frac{\sigma_1(\epsilon_c - \epsilon_s)}{P_S(\mathbf{x})} \right) \right] \\ &\approx E_{\mathbf{x} \sim P_C(\mathbf{x})} \left[\frac{\sigma_1(\epsilon_c - \epsilon_s)}{\ln 2 \cdot P_S(\mathbf{x})} \right], \\ KL[P_S \| P_C] &\approx E_{\mathbf{x} \sim P_S(\mathbf{x})} \left[\frac{\sigma_1(\epsilon_s - \epsilon_c)}{\ln 2 \cdot P_C(\mathbf{x})} \right]. \end{aligned} \quad (11)$$

From Eq. (11) above, it can be seen that the key factors that affect the KL-Divergence are the Gaussian vectors ϵ_s and ϵ_c used in the two types of generation processes and the parameter σ_1 .

For the former, the Gaussian vectors used in the two kinds of generation processes are of different origins: in the normal generation process, the vector ϵ_c comes from random sampling of the Gaussian distribution; in the message embedding process, the Gaussian vector ϵ_s is derived from the secret message of a uniform distribution through the message mapping method. As previously analyzed, since the secret message is encrypted by a computationally secure encryption algorithm, it can be considered computationally indistinguishable from true randomness. Therefore, the latter can also be regarded as obtained from random sampling from a Gaussian distribution. In this case, ϵ_c and ϵ_s can be considered to conform to the standard Gaussian distribution. In practice, since Gaussian noise is discretely sampled, its KL-Divergence tends to be 0.

The parameter σ_1 is specified in Eq. (5). It represents the magnitude of the Gaussian noise removed by the diffusion model in the denoising process and decreases gradually with the increasing number of denoising steps. The smaller the amplitude, the more refined the audio generation process. The generated cover and stego audio will be closer to the actual audio distribution. Predictably, the KL-Divergence also decreases with more denoising steps. Objectively speaking, the user of steganographic systems can choose the appropriate number of denoising steps according to their own needs, and make a trade-off and balance between system security and response speed, the latter of which grows with the increasing number of denoising steps as a matter of course.

In summary, the KL-Divergence values of the cover and stego audio of the CoAS system will be at a minimal positive value and decrease with the increasing number of denoising steps. This will be confirmed in the subsequent experiment Section IV-B.

2) *Security Specifications*: In Section III-A, we have defined the threat model of the generative steganography system. Under this threat model, we will discuss the security of the CoAS system and the diversity of generated results.

For the convenience of discussion, we simplify the description of the CoAS system. Consider the CoAS steganography system consisting of two parts, (G_1, G_2) . The audio **text** is generated by the initial **prompt** through the large language model (G_1) , $text_s = G_1(prompt, seed_1, seed_2)$, and the stego

audio is generated by the **text** through the diffusion models (G_2), FastDiff+ProDiff, $audio_s = G_2(text, seed_2, message)$.

In the CoAS steganography system, we used the provably secure linguistic steganography method, Discop, as an example, and the ϵ -secure audio steganography method proposed in this paper. Discop needs to use $seed_1$ to fix the sampling value to achieve the embedding of $seed_2$. The generation of the diffusion models requires $seed_2$ to fix the Gaussian noise sequence. It is worth noting that in the CoAS system, the sender does not need to choose the value of $seed_2$, which can be randomly selected by the model. According to our parameter synchronization technology in Section III-D, the randomly selected $seed_2$ can still be correctly extracted by the receiver and used in the message extraction process.

The attacker $\mathcal{A}$ has black-box access to the models (G_1, G_2) used by the CoAS system, and to the original models (U_1, U_2) for normally generated **audio** and **text** from the initial **prompt**. The attacker will conduct *chosen stego attacks* on the CoAS system: He can arbitrarily choose an initial **prompt**, repeatedly generate **text** and **audio** through two sets of models, and compare the results, trying to distinguish the differences.

The security of the CoAS system under this threat model can be summarized as the difference in the probability of distinguishing between normally generated results and the stegos. For any polynomial-time attacker $\mathcal{A}$, there is the following probability difference formula:

$$\left| Pr[\mathcal{A}_{(prompt)}^{G_1(p, s_1, s_2), G_2(t, s_2, \emptyset)}] - Pr[\mathcal{A}_{(prompt)}^{U_1(p, 1^{k_1}), U_2(t, 1^{k_2})}] \right| < \rho. \quad (12)$$

The 1^{k_1} and 1^{k_2} in the above formula represent the randomness in the normal generation process and are the source of diversity. Besides, s_2 is randomly selected, that is, $s_2 \leftarrow 1^{k_G}$.

Since the Discop is a provably secure linguistic steganography method, the following formula holds:

$$\left| Pr[\mathcal{A}_{(prompt)}^{G_1(p, s_1, s_2)}] - Pr[\mathcal{A}_{(prompt)}^{U_1(p, 1^{k_1})}] \right| \leq \epsilon_0, \quad (13)$$

where ϵ_0 is a negligible value under polynomial time conditions. At the same time, in Section III-E.1, we proved that the audio steganography based on the diffusion models in the CoSA system is ϵ -secure, so the following formula holds:

$$\left| Pr[\mathcal{A}_{(text)}^{G_2(t, s_2, \emptyset)}] - Pr[\mathcal{A}_{(text)}^{U_2(t, 1^{k_2})}] \right| \leq \epsilon_A. \quad (14)$$

We set the security parameter of the CoAS system to k . The length of the random number seed s_1 is k_s bits, $k_s = \text{Poly}(k)$. According to Discop, which is a provably secure linguistic steganography method, the probability that attacker $\mathcal{A}$ obtains s_1 from the generated text $G_1(p, s_1, s_2)$ is equal to random guessing. Combined with Eq. (13) (14), for ρ in Eq. (12), we have:

$$\begin{aligned} \rho &\leq 2^{-k_s} \times 1 + (1 - 2^{-k_s})(\epsilon_0 + \epsilon_A) \\ &= 2^{-k_s} + \epsilon_0 + \epsilon - 2^{-k_s}\epsilon_0 - 2^{-k_s}\epsilon_A \\ &= \epsilon + 2^{-k_s}(1 - \epsilon_A) + (1 - 2^{-k_s})\epsilon_0. \end{aligned} \quad (15)$$

In Eq. (15), since ϵ_0 is negligible, and k_s is a polynomial function of the security parameter k , the latter two terms $2^{-k_s}(1 - \epsilon_A)$ and $(1 - 2^{-k_s})\epsilon_0$ in the formula are negligible

compared with the security parameter k under polynomial time conditions.

In summary, the probability difference ρ in Eq. (12) satisfies the condition of $\rho \approx \epsilon$ under polynomial time conditions. The ϵ -security of the CoAS steganography system is proved. In addition, since the value of $s_2 \leftarrow 1^{k_G}$ is randomly selected by the models during the generation process of the CoAS system, this also preserves the diversity of text and audio generation.

IV. EXPERIMENTS

In this section, experimental results and analysis are presented to demonstrate the effectiveness and security of the CoAS steganography system.

A. Settings

First, we choose the FastDiff [33] and ProDiff [34] models as the audio generators, which have achieved significant results in the implementation of audio synthesis and generation. The dataset used for the experiments is LJSpeech [43], which consists of 13100 short audio clips consisting of a single speaker reading passages from 7 non-fiction books. The audio clips in the dataset are sampled at 22.05 kHz. We used 80-band mel-spectrograms as the conditions for speech synthesis. The *FFT_size*, *window_size*, and *hop_size* were set to 1024, 1024, and 256, respectively. Both models were run on NVIDIA RTX A6000 GPUs.

Following the audio generation and message embedding process described in Fig. 2, the random Gaussian distribution noise sequence and mel-spectrogram are used to generate the normal cover audio, whereas the Gaussian vector $\mathbf{z}$ obtained from the mapping of the encrypted message $\mathbf{m}$ is used to generate the stego audio. All audio segments are stored in uncompressed WAV format with a sampling rate of 22.05 kHz.

B. Steganographic Security

To demonstrate the ϵ -security of CoAS, we need to calculate the KL-Divergence between the cover and stego audio. Since the diffusion models cannot give an explicit probability distribution of the generated audio, we use frequency statistics to obtain the probability distribution of the audio sequence. We normalize the audio to $[-1, 1]$ and then divide the interval into a certain number of bins. The frequency distribution of the audio sequence values is regarded as a probability distribution, and the KL-Divergence between the two audio sequences is calculated.

We generated a series of cover and stego audio with 8 denoising steps and then computed the KL-Divergence. A total of 1000 audio groups corresponding to the same text are included. The number of bins selected when calculating is 64.

From Table I, it is clear that the KL-Divergence between the cover and stego audio is a very small value, which meets the definition of ϵ -security. Moreover, when the steganographic system randomly generates different cover audios with the same text, the KL-Divergence between them is close to the fluctuation during the steganographic process, which further

TABLE I
THE KL-DIVERGENCE FOR DIFFERENT AUDIO FILE PAIRS

Audio Pairs	KL-Divergence
<i>stego/cover</i> ₁	0.0514
<i>stego/cover</i> ₂	0.0533
<i>cover</i> ₁ / <i>cover</i> ₂	0.0586

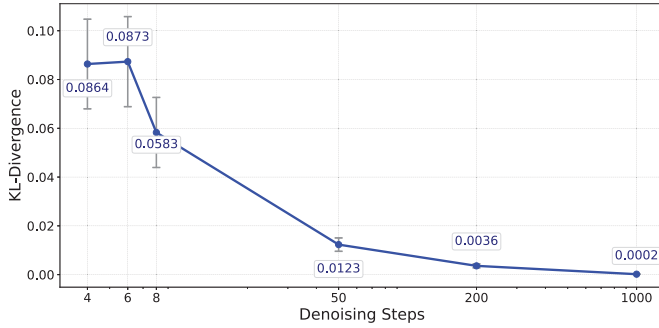


Fig. 5. KL-Divergence changes with the number of denoising steps.

supports the security and indistinguishability of the steganographic system.

Given our discussion on the ϵ -security of the CoAS system in Section III-E.1, the results under different denoising steps need to be considered when calculating the KL-Divergence. We select several different denoising steps and calculate the KL-Divergence values separately. For each denoising step, we generated 1000 pairs of cover and stego audio.

Under the same number of denoising steps, we selected several different numbers of bins to calculate the KL-Divergence, specifically 4, 8, 16, 32, 64, and 128. To optimize the appearance of the figure chart, we express the KL-Divergence results under different bins in the form of error bars.

It can be seen from the Fig. 5 that the value of KL-Divergence decreases and tends to zero with more denoising steps. This shows that the CoAS has some flexibility in terms of security. Users can adjust the number of denoising steps according to their needs, striking a balance between security and practicality. A lower number of denoising steps means faster generation speed, which we will explain in the following Section IV-C.

To verify that CoAS is fundamentally resistant to detection by existing steganalysis methods, we used the steganalysis networks Chen-Net [27] and Lin-Net [44] to directly detect the generated stego audios. Furthermore, we compared the accuracy with three different audio steganography methods, including the traditional LSB steganography and existing generative audio steganography [28] based on GAN. Each class of methods generated 1000 pairs of cover-stego audio for steganalysis. The metric used in the steganalysis experiment was the detection error rate $\bar{P}_E$, which is defined in the equation below:

$$\bar{P}_E = \frac{1}{2} (P_{FA} + P_{MD}), \quad (16)$$

TABLE II
THE ACCURACY OF STEGANALYSIS WITH DIFFERENT STEGANOGRAPHY ALGORITHMS

Steganography	Steganalysis	
	Chen-Net	Lin-Net
LSB [21]	20.16	17.64
Based on GAN [28]	44.51	42.39
CoAS	49.08	50.12

TABLE III
DIFFERENT DENOISING STEPS

Denoising steps	RTF↓	PESQ↑	STOI↑
5	0.053	2.51	0.84
50	0.78	2.55	0.84
200	1.31	2.76	0.89
1000	6.24	2.88	0.92

where P_{FA} denotes the false alarm rate and P_{MD} denotes the missed detection rate.

As shown in Table II, the result of CoAS is significantly better than those of the other methods, and the steganalysis methods fail to detect whether the audio file contains embedded secret messages. Therefore, our method can resist the detection of steganalysis methods.

C. Efficiency and Capacity

In the section above, we mentioned that the number of denoising steps (denoted as t) of the diffusion model determines the speed of audio generation. We measured the generation speed via the Real-Time Factor (RTF), which represents the time required to generate each second of audio. At the same time, we also measured the audio quality at different denoising steps. The experimental results are presented in Table III.

The results show that the RTF increases gradually with the number of denoising steps, but at the same time, the quality and intelligibility of the generated audio also improve. Therefore, users can choose different denoising steps based on their specific needs to strike a balance between audio quality and generation speed.

In addition to the generation speed of the actual audio generation process considered above, we also discuss the additional time cost caused by the embedding behavior of the secret message as one of the considerations for practical applications of the CoAS steganography system. According to Algorithm 1, the sender needs to traverse the entire secret message, perform the fragmentation and mapping process according to the payload p . Besides, according to Algorithm 2, the message extraction process of the receiver is the inverse process of the embedding process and has the same time cost. In general, the embedding and extraction behavior of the secret message introduces an additional $\mathcal{O}(n)$ time complexity.

TABLE IV

TIME CONSUMPTION DURING DIFFERENT GENERATION PROCESSES

Time (s)	Total Audio	Gaussian vectors (Ratio)
<i>cover</i>	33.0721	0.2031 (0.61%)
<i>stego</i>	33.2104	0.3372 (1.02%)

The experimental results in the Table IV verify the above inferences. When generating a pair of cover and stego audio, we not only count the total audio generation time, but also count the time consumed by the two processes to obtain Gaussian vectors ϵ_c and ϵ_s during the generation process. The former is randomly sampled from the Gaussian distribution by the FastDiff model, and the latter is obtained by the message mapping method from the secret message.

We generated an audio clip of about 5.4 seconds with 1000 denoising steps and embedded the secret message in the process of generating the stego audio. The above results show that the message mapping process of the secret message only increases the time cost of the sampled Gaussian vectors by about 0.41%, which is almost negligible as we predicted.

Considering the practical application of the CoAS steganography system, we will recommend the required GPU VRAM to users according to the experimental results (In fact, this is a necessary condition for running the audio diffusion model and the large language model). For example, when using the FastDiff and ProDiff models to generate an audio of about 15 words, the required GPU VRAM size is 2 GB. At the same time, to achieve steganographic parameter synchronization, we need to use a large language model to generate audio text, which is a considerable overhead. For example, running the LLaMA2-7B [38] model with FP16 precision to generate audio text requires about 24 GB of GPU VRAM overhead.

The capacity of the CoAS steganography system is determined by the main parameter payload p of the message mapping module, which means that the capacity of CoAS is exactly p bits per sample (bps). The upper bound of the capacity is the floating-point format used by the user during the audio generation process.

During the communication process described above, the capacity of Discop to hide the random number seed in the generated text is also a non-negligible part. It is directly related to whether an attacker can use an exhaustive method to crack the CoAS system. We experiment with the Discop [37]+SyncPool [39] using LLaMA-2 [38]. We introduce top- k sampling to constrain the size of the initial candidate pool and the probability distribution. To better evaluate the performance in various scenarios, we set the truncation parameter $k = 16, 32, 64, 128$. For each k , we select 100 initial contexts and generate the subsequent 50 tokens for each, averaging the results of capacity.

From Table V, it can be seen that the capacity of Discop is related to the value of k . When $k = 16$, the capacity of the random seed is about 1.13 bits per token. Given the CSPRNG algorithm used by the CoAS system, the maximum length of

TABLE V

THE CAPACITY OF DISCOP WITH LLAMA-2

k	16	32	64	128
Capacity (bits/token)	1.13	1.04	0.91	0.69

the random number seed is 32 bits. This also means that a text of at least 30 tokens is required.

During covert transmission through the CoAS system, the sender needs to generate an audio sequence of appropriate length based on the secret message. The length of the audio text is affected by two factors: on the one hand, when using provably secure linguistic steganography, the random number seed needs to be embedded in the audio text, and on the other hand, after the audio text is converted into audio sequence, there needs to be enough capacity to embed the secret message.

The former is directly related to the length s of the random number seed embedded by the sender. According to the embedding capacity of Discop in Table V, we round the embedding capacity to 1 bit per token, and the length of the audio text generated is at least s tokens.

As for the text length requirement in the audio steganography process, we assume that the payload is p . Under the default experimental conditions, a single English word is converted into mel-spectrograms with a time length of 15 using the ProDiff model, depending on the sum of the durations of the phonemes in the word. In the process of converting the mel-spectrograms into an audio sequence, the parameter *hop_size* of the FastDiff model is set to 256. This means that a single English word will generate an audio sequence of length $15 \times 256 = 3840$ on average. Therefore, for the sender, it is necessary to satisfy $3840p \cdot n_w > l_m$, that is, $n_w > \frac{l_m}{3840 \times p}$, where n_w represents the number of English words in audio text. When using large language models such as LLaMA-2 to generate audio text, each English word consists of an average of 3 tokens. The sender needs to set the maxtoken of the large language model to $\frac{l_m}{1240 \times p}$.

In summary, the sender needs to generate an audio text of at least $\max\left(s, \frac{l_m}{1240 \times p}\right)$ tokens to meet the requirements of embedding a random number seed using provably secure linguistic steganography and embedding secret messages via ϵ -secure audio steganography in CoAS.

D. Audio Quality

The audio steganography system described in this paper is based on diffusion models, which have achieved remarkable results in the field of audio generation. There are various evaluation metrics for the audio generated by the generative model, among which the most common are Perceptual Evaluation of Speech Quality (PESQ) and Short-Time Objective Intelligibility (STOI [45]). The PESQ, introduced by ITU-T in the P.862 standard, assesses the quality of audio by comparing it to the original audio through a series of transformations. The PESQ score ranges from -0.5 to 4.5 ,

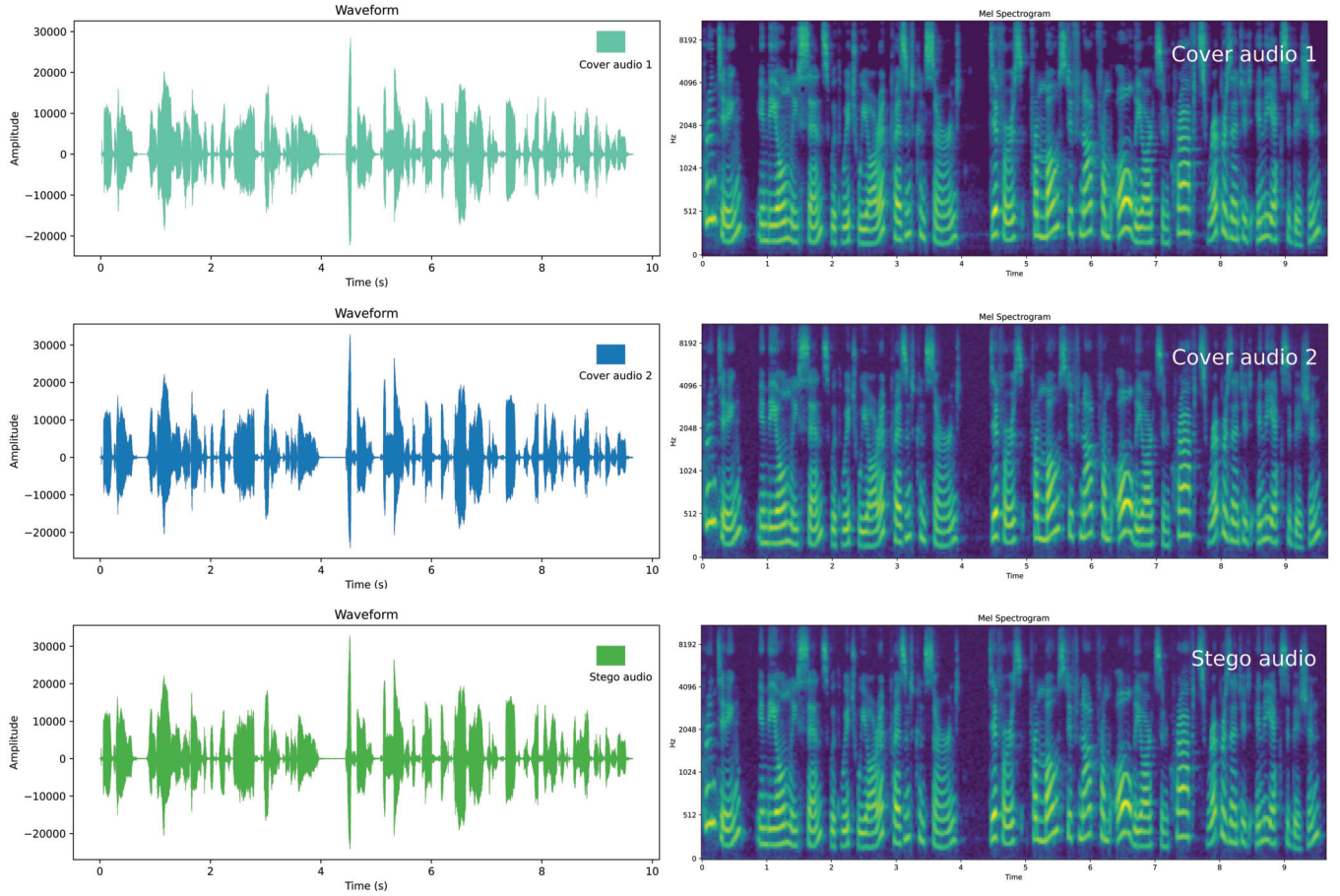


Fig. 6. The visualization of the cover audios triggered by different random seeds and the stego audio generated by diffusion stegosystem, including waveforms and mel-spectrograms. The waveforms and the mel-spectrograms are similar between the covers and the stego, indicating the stego audio has no special deviation from the cover.

with higher scores indicating better speech quality. The STOI, on the other hand, is an important metric for measuring speech intelligibility, representing the percentage of audio that can be fully understood by humans. A value closer to 1 indicates clearer and more intelligible audio. There is also an objectively assessed metric **Mel-Cepstral Distance (MCD [46])**, which is used to assess the compatibility between the spectra of two audio sequences.

In addition, a subjective indicator, **Mean Opinion Score (MOS [32])**, was chosen to measure the quality of the generated stego audio. It adopts five levels to evaluate the quality of the tested speech. Before the start of the evaluation, each listener was required to listen to a series of audio samples from the real dataset to ensure that they had the same standard as much as possible. We randomly selected 100 stego audio files for the experiment. Each listener listened to them and gave a score for each. Each stego audio received 6 scores from 6 different listeners. Finally, the average score of the stego audio is calculated.

We compared our method with several existing Text To Speech (TTS) methods in terms of audio quality, including Grad-TTS [47], Glow-TTS [48], GANSpeech [49], and DiffSpeech [50]. For sampling, we use 64 steps in Grad-TTS and 128 steps in DiffSpeech. For the FastDiff model in the CoAS, we choose the number of denoising steps to be 50. The

TABLE VI
THE AUDIO QUALITY

Method	PESQ $\uparrow$	STOI $\uparrow$	MCD $\downarrow$	MOS $\uparrow$
GT	/	/	/	4.39 $\pm$ 0.06
Grad-TTS [47]	1.48	0.19	3.36	4.05 $\pm$ 0.06
Glow-TTS [48]	1.48	0.19	4.35	4.01 $\pm$ 0.07
GANSpeech [49]	0.96	0.21	4.02	4.09 $\pm$ 0.06
DiffSpeech [50]	2.40	0.83	3.48	4.09 $\pm$ 0.06
FastDiff+ProDiff	2.56	0.85	3.15	4.08$\pm$0.07
CoAS (50 steps)	2.55	0.86	3.14	4.08 $\pm$ 0.05

GT means the ground-truth audio. For each audio generation method, we generated 1000 audio files and calculated the average value of quality. The results are shown in Table VI.

As shown in Table VI above, the diffusion models, FastDiff+ProDiff, for audio generation have made significant progress in the TTS task. It outperforms other non-autoregressive baseline models in terms of audio quality. The last two rows of data demonstrate the negligible impact of our audio steganography method, CoAS, on the quality of the diffusion model itself, achieving similar results across multiple

TABLE VII

THE SPEECH RECOGNITION ACCURACY WITH DIFFERENT TOOLS

Speech Recognition Models	WER↓	STOI
Canary-1B [52]	2.3%	0.85
Parakeet-TDT [53]	1.4%	
Whisper-large-v3 [54]	1.2%	

metrics. It also shows that our method achieves composite audio steganography while maintaining a high standard of audio quality and intelligibility.

E. Speech Recognition

In the CoAS system, a necessary step for the receiver is to obtain the same audio text as the sender through speech recognition, which is accomplished via speech recognition tools. We selected several existing excellent speech recognition models from the Huggingface Open ASR Leaderboard [51], which are deep learning models based on the transformer architecture.

We generated 1000 text-audio pairs and conducted speech recognition on them. We use Word Error Rate (WER) to measure the accuracy of the speech recognition system. It calculates the percentage of words in the system's output that differ from the correct audio text. The lower the WER value, the higher the accuracy. Additionally, the STOI [45] is an important indicator for measuring the clarity of speech and the degree of word understanding.

Table VII shows that under the existing speech recognition conditions, the receiver can recover the audio text with a high accuracy. In order to prevent the low probability of speech recognition errors from hindering the normal operation of the CoAS system, the sender can first use the speech recognition model to recognize the stego audio before transmitting it. Only send the stego audio after confirming that it can be 100% recognized; otherwise, repeat the steganography process of the CoAS with a different initial context. The high intelligibility also ensures that the audio will not arouse the suspicion of the attacker.

F. Visualization

Fig. 6 shows the waveforms and corresponding mel spectrograms of the cover and stego audio after conversion. For the same input text, a total of three audio segments were generated. Two segments were generated through the normal generation process, whereas the CoAS generated the last segment after message embedding. As can be seen, the three audio segments are similar and are not significantly different from each other. There is no presence of any specific characteristic that identifies a particular category.

V. CONCLUSION

In this paper, we discussed the shortcomings of existing traditional generative audio steganography methods: poor audio quality and limited diversity of audio generation. To address

the above issues, we proposed CoAS, an ϵ -secure composite audio steganography method based on text and speech synthesis. We use a provably secure linguistic steganography method to embed the synchronous side information required for audio steganography, and then replace the Gaussian noise in the diffusion models with message-driven sampling during the audio generation process. Numerous experimental results verify the ϵ -security and excellent audio quality of the CoAS. We would like to clarify the limitations and future work of the CoAS:

- To ensure that the message is fully extracted, we have made a slight modification to the normal diffusion model audio generation process, which could be the weakness of the CoAS. We plan to utilize more advanced generation methods to optimize this scheme in future work.
- Although the CoAS is ϵ -secure, the loss during audio transmission is not taken into account. Future research will improve the robustness of CoAS without affecting the steganography effect.

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